

Rでの解析の流れ

```
> 18-10A_gokamoku <- read.table('./seiseki_gokamoku.csv', header=T, sep=',')
> head(18-10A_gokamoku)
  ID kokugo suugaku eigo butsuri syakai
1  A     98      71   96      70     96
2  B     89      67   78      41     89
3  C     68      29   57      20     70
4  D     90      78   75      53     95
5  E     71      32   62      25     77
6  F     59      29   50      21     58

> 18-10A_gokamoku_fact <- factanal(18-10A_gokamoku[,2:6], factors=2, scores='regression', rotation='none')
> 18-10A_gokamoku_fact

Call:
factanal(x = df[, data_cols], factors = factors, scores = scores,      rotation = rotation)

Uniquenesses:
kokugo suugaku      eigo butsuri  syakai
0.005   0.067   0.066   0.005   0.057

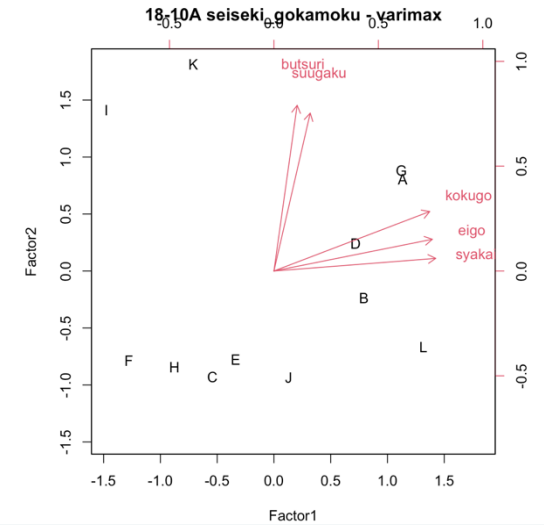
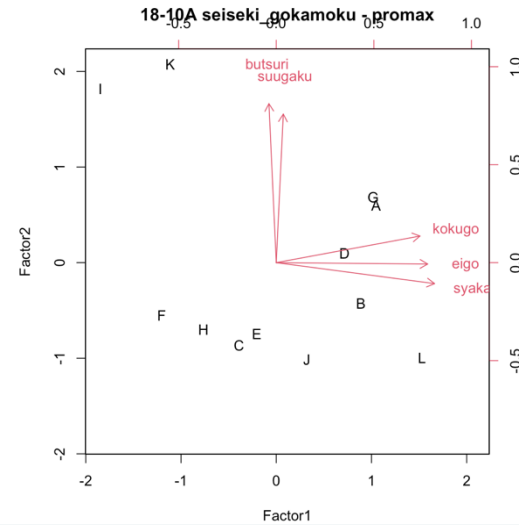
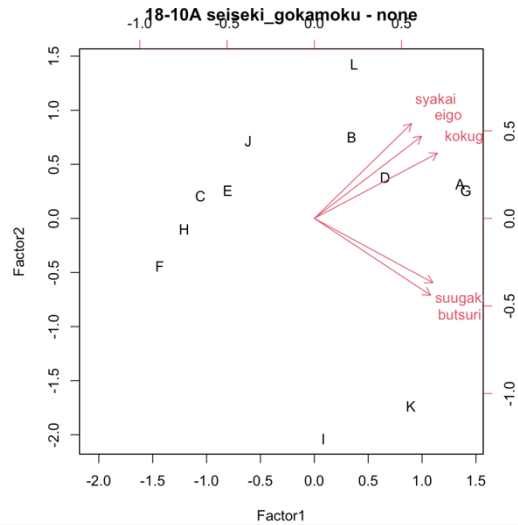
Loadings:
      Factor1 Factor2
kokugo  0.882   0.465
suugaku 0.850  -0.459
eigo    0.768   0.586
butsuri 0.833  -0.548
syakai  0.696   0.677

      Factor1 Factor2
SS loadings  3.270   1.530
Proportion Var 0.654   0.306
Cumulative Var 0.654   0.960

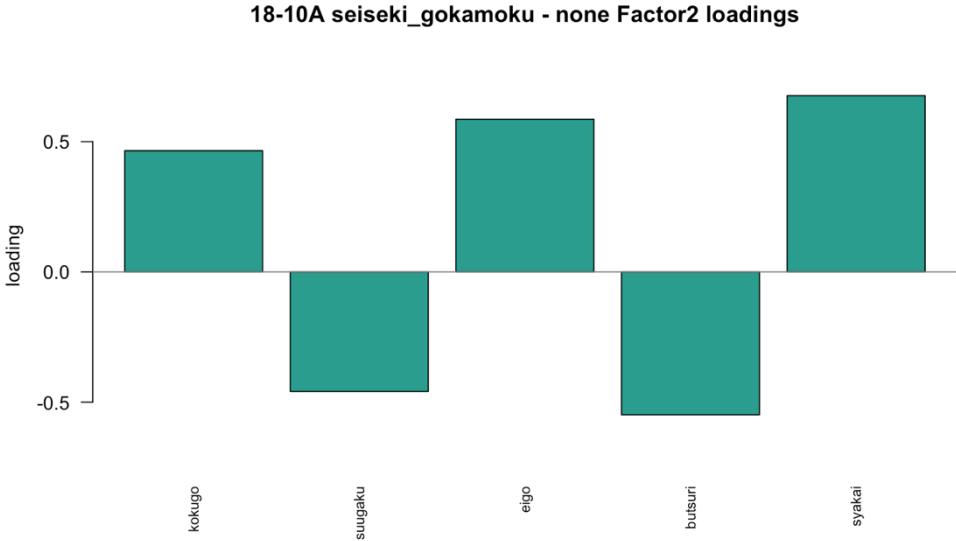
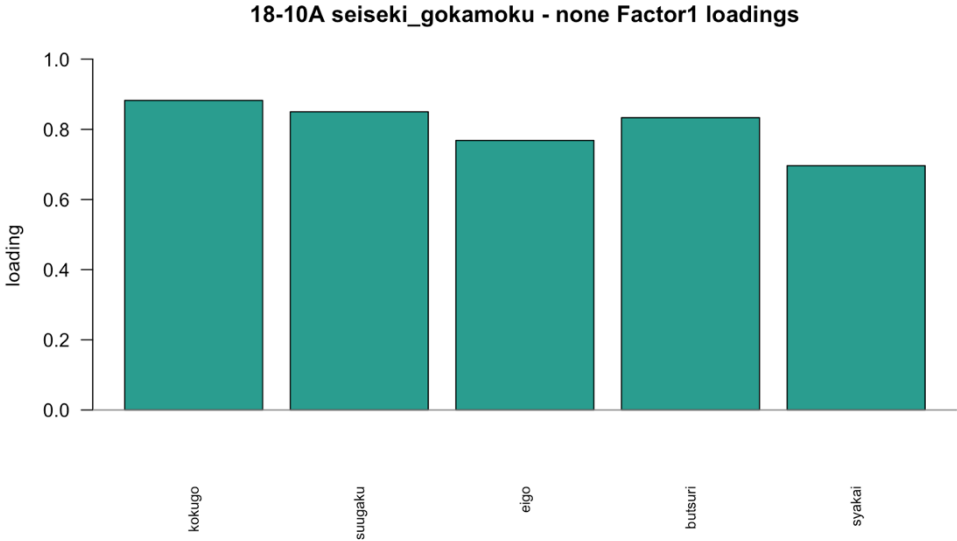
Test of the hypothesis that 2 factors are sufficient.
The chi square statistic is 3.45 on 1 degree of freedom.
The p-value is 0.0634

> 18-10A_gokamoku_fact$loadings[,1:2]
      Factor1 Factor2
kokugo 0.8824152 0.4652646
suugaku 0.8499213 -0.4589035
eigo    0.7682944 0.5863149
butsuri 0.8332433 -0.5483724
```

因子得点と因子負荷量



因子負荷量の棒グラフ



ソースコード (1/3)

Class13_rp18-10A_02.R

```
1 setwd("/Users/kakigoori/Desktop/data/13")
2
3 dir.create("./answers/plots_pdf", recursive=TRUE, showWarnings=FALSE)
4
5 gokamoku <- read.table("./seiseki_gokamoku.csv", header=T, sep=",")
6
7 gokamoku
8
9 head(gokamoku)
10
11 gokamoku_fact <- factanal(gokamoku[,-1], factors=2, scores="regression", rotation="none")
12
13 str(gokamoku_fact)
14
15 gokamoku_fact
16
17 gokamoku_fact$loadings[1,]
18
19 gokamoku_fact$loadings[1,]^2
20
21 1 - sum(gokamoku_fact$loadings[1,]^2)
22
23 apply(gokamoku_fact$loadings, 1, function(x){1-sum(x^2)})
24
```

ソースコード (2/3)

Class13_rp18-10A_02.R

```
25 gokamoku_fact$scores
26
27 pdf("./answers/plots_pdf/18-10A_gokamoku_none_biplot.pdf", width=8.5, height=6.2)
28 biplot(gokamoku_fact$scores, gokamoku_fact$loadings, xlab=gokamoku[,1])
29 dev.off()
30
31 pdf("./answers/plots_pdf/18-10A_gokamoku_none_factor1_bar.pdf", width=8, height=5.2)
32 barplot(gokamoku_fact$loadings[,1], cex.names=0.8, las=2, ylim=c(0,1))
33 abline(h=0, col="gray50")
34 dev.off()
35
36 pdf("./answers/plots_pdf/18-10A_gokamoku_none_factor2_bar.pdf", width=8, height=5.2)
37 barplot(gokamoku_fact$loadings[,2], cex.names=0.8, las=2, ylim=c(-0.8,0.8))
38 abline(h=0, col="gray50")
39 dev.off()
40
41 gokamoku_fact <- factanal(gokamoku[, -1], factors=2, scores="regression", rotation="promax")
42
43 pdf("./answers/plots_pdf/18-10A_gokamoku_promax_biplot.pdf", width=8.5, height=6.2)
44 biplot(gokamoku_fact$scores, gokamoku_fact$loadings, xlab=gokamoku[,1])
45 dev.off()
46
47 pdf("./answers/plots_pdf/18-10A_gokamoku_promax_factor1_bar.pdf", width=8, height=5.2)
48 barplot(gokamoku_fact$loadings[,1], cex.names=0.8, las=2, ylim=c(-0.3,1.2))
```

ソースコード (3/3)

Class13_rp18-10A_02.R

```
49 abline(h=0, col="gray50")
50 dev.off()
51
52 pdf("./answers/plots_pdf/18-10A_gokamoku_promax_factor2_bar.pdf", width=8, height=5.2)
53 barplot(gokamoku_fact$loadings[,2], cex.names=0.8, las=2, ylim=c(-0.3,1.2))
54 abline(h=0, col="gray50")
55 dev.off()
56
57 gokamoku_fact <- factanal(gokamoku[, -1], factors=2, scores="regression", rotation="varimax")
58
59 pdf("./answers/plots_pdf/18-10A_gokamoku_varimax_biplot.pdf", width=8.5, height=6.2)
60 biplot(gokamoku_fact$scores, gokamoku_fact$loadings, xlab=gokamoku[,1])
61 dev.off()
62
63 pdf("./answers/plots_pdf/18-10A_gokamoku_varimax_factor1_bar.pdf", width=8, height=5.2)
64 barplot(gokamoku_fact$loadings[,1], cex.names=0.8, las=2, ylim=c(-0.3,1.2))
65 abline(h=0, col="gray50")
66 dev.off()
67
68 pdf("./answers/plots_pdf/18-10A_gokamoku_varimax_factor2_bar.pdf", width=8, height=5.2)
69 barplot(gokamoku_fact$loadings[,2], cex.names=0.8, las=2, ylim=c(-0.3,1.2))
70 abline(h=0, col="gray50")
71 dev.off()
```